



2025

J24

BOOKLET NO.

102081

Forest Services
Civil Engineering

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

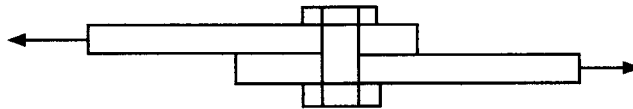


SECTION – A

Q1. Solve **any five** out of seven. **(8×5=40)**

Q1. (a) i) Define angular velocity and angular acceleration. State their relation. **8**
ii) State D'Alembert's principle and explain its significance in dynamics.

Q1. (b) i) Calculate bolt value of 20 mm dia bolt connecting two plates of 10 mm for 4.6 grade bolt and Fe410 grade plate (Figure given below). **8**



ii) Two plate of size 200×10 mm are connected by lap joint using shop welding. Plates are Fe410 grade. Design a fillet weld of full strength of plate.

Q1. (c) i) What are the properties of fluid ? Explain their role in fluid motion. **8**
ii) Explain the terms, velocity potential function and stream function.

Q1. (d) i) Explain the classification of soil as per IS classification system. **8**
ii) Explain the various factors affecting permeability.

Q1. (e) i) Differentiate between three-hinged and two-hinged arches. **8**
ii) Define stiffness factor and carry-over factor.

Q1. (f) i) Explain methods and systems of prestressing. **8**
ii) Explain in details losses in prestress concrete as per IS code.

Q1. (g) i) A jet of water of diameter 50 mm moving with a velocity of 40 m/s, strikes a curved fixed symmetrical plate at the centre. Find the force exerted by the jet of water in the direction of the jet. If the jet is deflected through an angle of 120° at the outlet of the curved plate. **8**

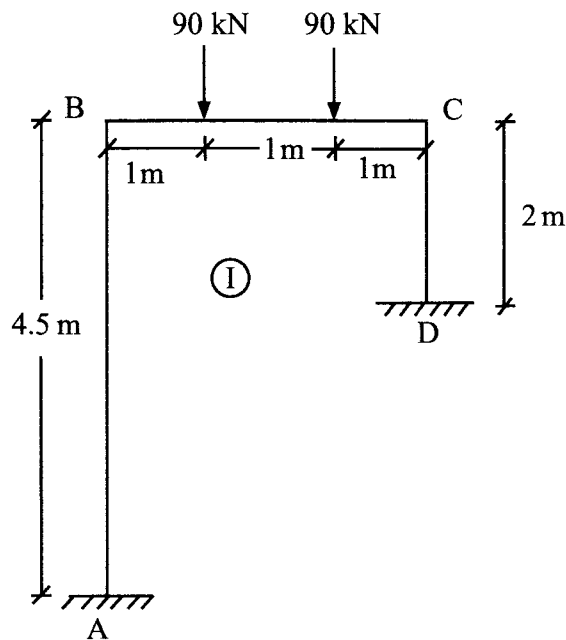
ii) What are Hydraulic Grade Line and the Total Energy Line ? Explain.



- Q2. (a) i)** Define stress and strain. Derive the relationship between Young's modulus, Bulk modulus and Poisson's ratio. **15**
- ii)** Derive expressions for hoop stress and longitudinal stress in thin cylinders.
- Q2. (b) i)** Design two-way slab of clear size $3 \text{ m} \times 4.5 \text{ m}$. Live load = 2.5 kN/m^2 . Slab is simply supported on four side with corner free to lift side support beam width is 225 mm. Use M15 Fe415. **15**
- ii)** Use limit state method to design a reinforced concrete beam for flexure only to support a uniformly distributed services load of 25 kN/m , over an effective simply supported span of 6 m. The material used are M20 grade concrete and HYSD steel of grade Fe415, Assume $d = 2b$.
- Q2. (c) i)** Explain the terms rapidly varied flow and gradually varied flow. **10**
- ii)** The specific energy for a 5 m wide rectangular channel is to be 4 Nm/N. If the rate of flow of water through the channel is $20 \text{ m}^3/\text{s}$, determine the alternate depth of flow.
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- Q3. (a) i)** Define the specific speed of a centrifugal pump. Derive an expression for the same. **15**
- ii)** What is a reciprocating pump ? Describe the principle and working of reciprocating pump with a neat sketch. Why reciprocating pump not coupled directly to the motor ?
- Q3. (b) i)** Enlist the various shear tests used to determine the shear strength parameters and explain in detail the vane shear test. **15**
- ii)** Determine the safe bearing capacity of square footing using IS code method. If the footing is resting at 1.5 m below G.L. The size of footing is 2.4 m and resultant load is inclined at 9° with vertical. Take $\gamma = 18 \text{ kN/cm}^3$; $c = 15 \text{ kN/m}^2$, $\phi = 30^\circ$, Take FOS = 2, $N_c = 30.14$, $N_q = 18.4$, $N_\gamma = 22.4$.
- Q3. (c) i)** Explain in detail sand drain method of ground improvement. **10**
- ii)** Explain in detail the pile load test.
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Q4. (a) Analyze the given frame by using Moment Distribution Method (Figure given below). 15



Q4. (b) Design a laced column with two channel back to back of length 10 m to carry an axial factored load of 1400 kN. The column may be assumed to have restrained in position but not in direction at both end (hinged ends). 15

Q4. (c) A vertical sluice gate is used to cover an opening in a dam. The opening is 2 m wide and 1.2 m high on the upstream of the gate, the liquid of sp.gr. 1.45 lies upto a height of 1.5 m above the top of the gate, whereas on the downstream side, the water is available upto a height touching the top of the gate. Find the resultant force acting on the gate and the position of the centre of pressure. Find also the force acting horizontally at the top of the gate which is capable of opening it. Assume that the gate is hinged at the bottom. 10

**SECTION – B**

Q5. Solve **any five** out of seven.

(8×5=40)

Q5. (a) State the BIS requirements for following construction materials.

8

i) Bricks

ii) Cement.

Q5. (b) Explain in brief the method of radiation and intersection used in plane table surveying.

8

Q5. (c) Explain the rational method used for estimating maximum flood discharge obtained from a catchment.

8

Q5. (d) The population census data of a town is given below. The rate of water demand is 160 Lpcd and the design period for the proposed water supply scheme is 30 years.

Census data

Year	1981	1991	2001	2011	2021
Population	21500	26400	32800	40200	50300

The water supply scheme will commence to operate from year 2026. Determine the water demand (requirement) for the design year 2056.

8

Q5. (e) Explain the principle, working and advantages of concrete batching plant for mass concrete works.

8

Q5. (f) Explain origin and destination study. What are the various uses of O and D study ?

8

Q5. (g) What is meant by canal alignment ? State the different methods and explain any one method.

8



Q6. (a) Write short notes on :

i) Methods for cost control and cost reduction for a construction project. 8

ii) Benefit-cost analysis for a construction project. 7

(8+7=15)

Q6. (b) Write short notes on :

i) Sleepers

ii) Ballast

iii) Overtaking sight distance

iv) Superelevation

v) Cross-sectional details of a highway in embankment. (3×5=15)

Q6. (c) Differentiate between PERT network and CPM network with reference to a suitable example of a project. (5+5=10)

Q7. (a) Explain in brief :

i) Method of computing design capacity of a canal. 7

ii) Describe the necessity and functioning of distributary head regulator and cross regulator in a canal project. (4+4=8)

Q7. (b) Write short notes on :

i) Explain the mass curve method used for determination of balancing reserve of a storage reservoir. 8

ii) State the different methods of distribution layout for water distribution network and explain any one method with advantages and disadvantages. (2+5=7)

Q7. (c) i) Differentiate between primary and secondary treatment of sewage. 4

ii) Explain the principle and working of activated sludge process using flow diagram. 6



Q8. (a) Explain in brief the following :

- i) Differentiate between flexible pavement and rigid pavement considering major advantages and disadvantages of each case. **10**
- ii) Write a short note on temporary adjustment of a theodolite. **5**

Q8. (b) Draw the flow diagram of a conventional water treatment for treatment of surface water from a dam and explain in brief principle and working of

- i) Rapid mix unit
- ii) Clariflocculator
- iii) Rapid sand filter. **(3+4+4+4=15)**

Q8. (c) Write short notes on :

- i) Gravity settling chamber
 - ii) Ambient air quality standards for PM₁₀, PM_{2.5}, SO₂ and NO₂. **(4+6=10)**
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